

CS 4632 Summer Section W01

Project Foundation

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I. BACKGROUND

THIS PROJECT started as an idea while working in dealership service departments. During the nearly six years, five dealerships, and four brands, I noticed multiple bottlenecks in the department. The bottlenecks seemed to be large in technician scheduling, parts availability, repair bay utilization, and customer intake. In a place where changing the existing workflow could cost the company significant resources, having a simulation program will assist decision-making.

A. Project Overview

1) *Name:* Service Department Operational Optimization Simulator

2) *Domain:* This project simulates the daily operation of an automotive dealership service department using a discrete-event simulation. Rather than modeling the physics behind vehicle dynamics, the simulation being developed focuses on operational workflows. This model looks at the relationship between customers, service advisors, technicians/repair(work) bays, and the parts department. The simulation outputs various metrics to identify bottlenecks, evaluate staffing decisions, and estimate the impact of the parts supply chain on shop performance.

3) *Problem Statement:* The key objective of this project is to analyze key metrics to find bottlenecks that increase customer wait times and decrease customer satisfaction and efficiency through reduced technician and bay utilization and productivity.

4) *Scope:* As the project is designed for an 8-week course, the scope will be limited. The core of the project will focus on developing the availability of parts, service advisor workflows, and technician workflows for a 30-bay service department. As customer payments do not have a major impact on overall workflow efficiency, they will not be included in this project's scope. Vehicle physics is too complex a task for the scope of this project.

II. PROJECT EVALUATION FRAMEWORK

A. Framework

1) *Modeling and Implementation:* This project is designed to implement mathematical equations in a simulation. As copying a dealership workflow is too simple for this project, there needs to be a form of metrics analysis to make this system more valuable to service department management and as a learning tool. Through the use of tracking several metrics

during the simulation, such as customer wait times, technician usage, repair bay usage, parts availability, and overall shop throughput, a system can be developed to measure the impact of operational decisions on the dealership. With the use of the stated metrics, a more accurate evaluation can happen [1]. Any hidden bottlenecks can be found by refining each metric. Adding features to future versions can link financial impacts to underutilized resources and delayed repairs.

2) *Uncertainty and Randomness:* Dealership Service Centers have a certain amount of randomness and uncertainty that can be simulated within this project [2]. As a key example, Customer arrivals are a random process that can be simulated by a mathematical equation [3]. Uncertainty shows up in multiple aspects, but this simulation will be limited by the scope. The ways uncertainty will be simulated are: the vehicle condition on arrival, which influences the amount and length of work needed; technician speed, which is primarily based on experience and the repair [4]; and parts availability, which relies on inventory and the supply chain [1].

3) *Interesting Interactions:* Considering that a dealership's departments are tightly coupled, an issue that arises in one department can cause slowdowns in another. A prime example is that a parts shortage or the wrong parts getting allocated to the ticket can cause technicians to wait even when repair bays remain available. This is expensive as warehouse storage costs increase, and technician utilization plummets. Similarly, increasing the headcount for technicians may not allow for greater utilization when the limiting factor is the advisors' ability to intake a customer. Creating a working model to visualize these interactions allows the relevant parties to see the bottlenecks in the department [1], [2].

B. Assumptions

As this is a condensed course to meet an 8-week schedule, several simplifications have been made to ensure the scope is sufficiently covered.

1) *System Components:* Using personal experience and a bit of research to develop the current model covered the properties to be robust enough to develop.

2) *Arrival Process:* As Customer arrivals are inherently complex to simulate, the scope is limited. Walk-in customers and those with appointments are placed in the same stream. No-shows and those with canceled appointments are not modeled due to time limitations. The last assumption in this category is that the arrival stream follows a Poisson process.

3) *Service Advisors:* During the intake process, a key simplification is that a service advisor's experience will not affect the time it takes to intake a customer.

4) *Technicians*: A key assumption is that all technicians are capable of performing every job that comes in [4]. Any experience gained during the simulation will not transfer to the next simulation. Various factors, such as fatigue and breaks, are also not within the scope of this project.

5) *Repair Bays*: To simplify the workflows, each technician will be allocated to only one bay. Specialized equipment is not specific to one bay. To allow a more focused scope, the bay capacity will remain constant during the simulation.

6) *Parts and Inventory*: For a more focused scope, the parts department will not sell parts to customers over the counter. Parts will go from the supplier to the on-site warehouse to the technician. Part orders that are high priority are fulfilled with a stochastic lead time from a singular supplier [1]. Outside of the lead time for each part, prices and supplier variability will not be considered within the scope of this project.

7) *Job Queue*: As all technicians are deemed to be competent and creative enough to perform every job, the Queue system will operate on a strict, single First in First out basis. There will also be no separate queue for customers who are dropping vehicles off, high-priority customers, or warranty repairs planned at the start of the project.

8) *Financial Modeling*: No financial modeling is planned at the start of the project; however is a possible extension to the project.

9) *Physical Environment*: Any sort of physics or dynamics-based simulation is outside the scope of the simulation. This simulation's scope is workflows and resource allocation only.

III. SYSTEM COMPONENTS AND DESCRIPTION

A. Customer

Each customer entity represents one vehicle arriving at the dealership. Each customer has several properties, which are arrival time, service advisor, service ticket, and average ticket processing time.

B. Service Advisor

Each Service Advisor entity intakes customers, creates a service ticket, and coordinates between the customer and technician. Based on experience, service advisors also build relationships with repeat customers and reduce customer intake times, but due to limitations, experience and relationships will not be stated properties. Properties include the current ticket queue, availability status, and average ticket processing time.

C. Service Ticket

A service ticket entity is generated by a service advisor during the intake process, and drives the entire workflow of the rest of the workflow. This class's properties are job type, estimated labor time, required part, assigned technician, and current status.

D. Technician

A technician entity performs the physical work on the vehicle and works with the parts department to get the correct parts to the correct vehicle. The properties for this class are skill level, current job assignment, utilization rate, bay assignment, and service time distribution [4].

E. Repair Bay

Repair bays are the workspaces assigned to the technicians. Class properties are occupancy status, assigned technician, and current vehicle.

F. Parts Department

The parts department fulfills parts requests created by the service ticket, handles supplier relationships, and manages parts inventory. Properties are current inventory levels, pending orders, and average fulfillment time [1].

G. Parts Inventory

A part entity is handled within the parts department and gets delivered to the technician. Properties are part number, quantity on hand, reorder threshold, and lead time.

H. Job Queue

The job queue holds service tickets awaiting technicians to become available. Based on the description and model, the properties are queue length, average wait time, and ticket priority.

IV. CORE MODELS AND ALGORITHMS

A. M/M/c Queuing Model

A big challenge in a service department is balancing incoming work with service capacity. Vehicles arrive and leave throughout the day, but the number of technicians and repair bays remains the same. As demand outpaces the capacity, the queue starts to increase. This causes an increase in customer wait times and hurts the overall efficiency metric. To model this behavior, the simulation uses a multi-server model or M/M/c queue, being an extension of the single-server model, allowing multiple technicians to work from a shared single queue [2], [5].

Using this model, customer arrivals are modeled using a Poisson arrival rate equal to λ , while service times are modeled using an exponential distribution with a service rate of μ . The variable c equals the number of technicians and service bays available. Seeing as how the author worked in multiple dealerships where service tickets were generally processed in a First-In First-Out (FIFO) queue, the simulation will reflect a FIFO-only system. This is primarily done to keep the simulated process as close to the intended process.

One of the main metrics within the system is system utilization- equal to ρ . System utilization is a measure of how much the available workforce is being used. It is calculated with:

$$\rho = \frac{\lambda}{c\mu} \quad (1)$$

As the system must remain stable with $\rho < 1$, customer arrivals must remain below the available service capacity. As utilization approaches 100%, queue lengths and customer wait times increase rapidly.

Using the formula- called the Erlang C formula- [6] shown below, the probability of an arriving customer waiting before work can begin can be calculated:

$$C(c, \lambda/\mu) = \frac{\frac{(c\rho)^c}{c!(1-\rho)}}{\sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{(c\rho)^c}{c!(1-\rho)}} \quad (2)$$

To better understand the system, a scenario will be used. The assumptions in this scenario will be $\lambda = 4$ vehicles per hour, a mean service time of $1/\mu = 0.5$ hours per technician, and $c = 3$ technicians on shift. The utilization will be calculated as:

$$\rho = \frac{4}{3 \times 2} = 0.667 \quad (3)$$

With these assumptions, the service department operates at approximately a 67% utilization. Then, the Erlang C formula will be used to estimate the probability that an arriving vehicle must wait before service begins. From that value, the expected queue wait time can be calculated as:

$$W_q = \frac{C(c, \lambda/\mu)}{c\mu - \lambda} \quad (4)$$

These metrics provide useful information for estimating shop efficiency and customer wait times. While the simulation is a discrete-event model, the M/M/c framework is an analytical tool that can be used to benchmark and validate simulation behavior. The numerical values shown above are intended only to illustrate the model structure; each dealership that uses this software will have parameters finely-tuned to its data.

B. Poisson Arrival Process

As customer arrivals are not a constant variable, they must be modeled using an equation. As most dealerships show predictable surges during specific time frames, mapping the behavior to a model can be relatively simple. To capture this behavior, customer arrivals are modeled using a non-stationary doubly stochastic Poisson process- more simply known as a Cox Process, where the arrival rate, λ , varies over time [3]. Using this approach provides the simulation access to represent realistic fluctuations in workload while keeping some level of randomness in customer arrivals. The probability of having n arrivals during a time t is:

$$P(N(t) = n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!} \quad (5)$$

Using time as a variable during the arrival process helps create a simulation more closely aligned with real-life operations. It is particularly visible when evaluating filling technician and service advisor roles, identifying periods where demand exceeds capacity, and low-demand hours.

C. Experience-Based Service Time Model

Each technician gets an experience level $e_i \in [0, 1]$ - zero being a new hire, one being at the master technician level [4]. That value feeds into a scaling function that shrinks their mean service time, shown as:

$$f(e_i) = \frac{1}{1 + \alpha \cdot e_i} \quad (6)$$

α is a tunable coefficient. Bigger α means experience buys more speed. The effective mean for technician i on a job with base duration μ_0 becomes:

$$\mu_i = \mu_0 \cdot f(e_i) \quad (7)$$

As service time is not deterministic due to multiple factors, it is drawn from a Gamma distribution:

$$T_i \sim \gamma(k, \theta_i), \quad \theta_i = \frac{\mu_i}{k} \quad (8)$$

The shape parameter k handles variance. Having a high k value means the technician is consistent; low k values mean their times vary more. Either way, the mean stays anchored to μ_i :

$$E[T_i] = \mu_i \quad (9)$$

$$\text{Var}(T_i) = \frac{\mu_i^2}{k} \quad (10)$$

The Gamma was chosen over the exponential specifically because it lets the variance and mean be tuned independently - something that matters when modeling the real difference between a technician who has performed a job a hundred times versus someone seeing it for the first time. k will be calibrated against actual dealership data before the simulation runs.

D. Inventory and Parts Availability Model

Parts availability is modeled using a stochastic inventory system [1]. Each part has an associated demand rate, reorder point r , and reorder quantity Q . When on-hand inventory falls below r , a replenishment order is triggered with a stochastic lead time L . A job requiring an unavailable part enters a blocked state, contributing to the total job delay:

$$D_{total} = D_{queue} + D_{parts} \quad (11)$$

where D_{queue} is the wait time before technician assignment and D_{parts} is the additional delay incurred while awaiting part fulfillment.

E. System Dynamics

The simulation models a continuous flow of vehicles through the dealership service department. The system progresses through four sequential stages: customer arrival, ticket creation, repair execution, and job completion.

1) *Customer Arrival*: Customers arrive according to a non-stationary Poisson process [3]. Upon arrival, the customer is assigned to an available service advisor. If no advisor is available, the customer waits in the advisor queue.

2) *Ticket Creation*: The service advisor intakes the customer, diagnoses the vehicle, and generates a service ticket. Vehicle condition on arrival introduces uncertainty at this stage, as the scope of work may differ from the initially reported issue. The completed ticket is placed into the technician job queue.

3) *Technician Assignment*: Technicians pull tickets from the shared job queue on a FIFO basis. If all technicians are occupied, the ticket waits in queue, contributing to D_{queue} . Upon assignment, the technician checks parts availability before beginning work.

4) *Parts Request and Fulfillment*: If required parts are in stock, the technician proceeds immediately. If parts are unavailable, the job enters a blocked state and the technician may pull another ticket from the queue. The blocked job resumes once parts are fulfilled, contributing D_{parts} to the total job delay [1]. This interaction directly affects technician utilization, per-bay throughput, and overall shop throughput.

5) *Repair Execution*: The technician completes the repair over a service time drawn from an experience-based distribution [4]. Upon completion, the bay is released and the technician returns to the job queue.

6) *Job Completion*: The completed ticket is returned to the service advisor, who notifies the customer. Total customer wait time is recorded as:

$$W_{total} = W_{advisor} + D_{queue} + D_{parts} + T_{service} \quad (12)$$

where $W_{advisor}$ is the advisor intake wait, D_{queue} is the technician queue delay, D_{parts} is the parts fulfillment delay, and $T_{service}$ is the active repair time.

V. IMPLEMENTATION

A. Language Choice

Based on academic and personal experiences, Java is the language of choice. Since most of the components can be mapped to classes, a language that supports object-oriented programming would be preferred. Since there are quite a few parts of a dealership that can be added at a later point, having Java's scalability would benefit the project significantly. Due to library support and experience, Java 22 and newer is preferred.

B. Environment

Development will be performed using JetBrains IntelliJ IDEA, JavaFX for visualization and user interaction, and SimKit for discrete-event simulation support. Source control and project documentation will be maintained through GitHub and Overleaf. AI-assisted development tools will be used throughout the implementation of this project. For documentation, ChatGPT is used for brainstorming [7] and assisting in generating citations aligned with IEEE standards from articles previously noted as being useful. Citation quality was verified by humans to see that all relevant information has been provided.

C. Simulation Type

The simulation will be implemented as a discrete-event simulation (DES). This approach is appropriate for modeling a service department, where system state changes are driven by distinct events such as customer arrivals, job assignments, parts requests, and job completions [2].

D. Tracked Metrics

The following performance metrics will be tracked throughout the simulation:

- Customer wait times
- Technician utilization
- Wait times for parts
- Jobs completed per day, reported both as a shop total and broken down per technician
- Job delays attributed to parts availability

VI. EXPECTED OUTCOMES

This project specifically has a few outcomes that would be expected based on experiencing several dealerships. A service department with more technicians and service bays will increase the throughput and reduce wait times, but having more technicians past the advisor intake capacity will do nothing outside of increasing costs. A second outcome that is intuitive is that part delays reduce utilization, though phantom under-utilization- technician has a vehicle in the bay, but no work is being completed- is a tradeoff that needs to be considered when simulating the parts department. The last major expected outcome will be regarding customer rushes. As experience dictates, the first morning rush causes the first queue and tends to be the longest. Because most dealerships run shifts similar to 7 am to 4 pm and 9 am to 6 pm, the 7 am shift tends to have less staffing, as the queue backlog from the morning customer rush tends to take the longest to empty out.

APPENDIX A SIMULATION PARAMETERS

TABLE I
INITIAL SIMULATION INPUT PARAMETERS

Parameter	Symbol	Value
Customer arrival rate	λ	TBD (Cox Process)
Mean service time per technician	$1/\mu$	TBD (Gamma)
Number of technicians/bays	c	TBD (calibrated from data)
Number of service advisors	-	TBD (calibrated from data)
Technician experience level	e_i	[0, 1]; calibrated per technician
Experience coefficient	α	TBD (calibrated from data)
Gamma shape parameter	k	TBD (calibrated from data)
Parts reorder point	r	TBD (calibrated from data)
Parts reorder quantity	Q	TBD (calibrated from data)
Parts lead time	L	TBD (Stochastic)
Simulation run length	-	10 hours (1 work day)
Number of replications	-	TBD (calibrated from data)

Variables marked "TBD" will be tuned at a later point.

APPENDIX B UML DIAGRAMS

UML diagrams for this project include a Class Diagram and an Activity Diagram, both designed in PlantUML. The Class Diagram illustrates the primary simulation entities - SimulationEngine, Customer, ServiceAdvisor, RepairTicket, Technician, RepairBay, PartsInventory, Part, JobQueue, and MetricsCollector - along with their attributes, methods, and relationships. The Activity Diagram captures the end-to-end simulation flow across four swimlanes: Customer, Service Advisor, Technician, and Parts Department. Although they are not included within this document, the class diagram and activity diagram have been attached in the same upload folder.

APPENDIX C GITHUB REPOSITORY

To maintain software best practices, the project will have frequent pushes to a public GitHub repository. As the repository will be publicly maintained, a Readme markdown file and gitignore file will be maintained. Keeping with the project requirements, the project structure has been prepared with a description of the project within a Readme file.

Repository URL: [GitHub Repository](#)

REFERENCES

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